v2 results review (24 Oct 2026)

Inputs frozen at git tag `v2-preanalysis` (871d199). Scripts 29-36 run 24 Oct 2026; two code-only fixes after the freeze (power-simulation overflow; bootstrap speed, coxph.fit with coxph's inputs) - numbers unchanged. Checks: scripts 33/34 reproduce the script 30/32 point estimates exactly (max difference 0); primary bootstrap 2,000 x 26, no failed replicates, replicate log-HRs -0.95 to 1.20; 5 non-convergence warnings among ~200,000 sensitivity bootstrap fits.

Headline results (v2)

- Positive control: all 7 cohorts pass. GSE31210/GSE50081/GSE30219 by the D1 rule (HR II vs I); GSE30219 rests on 3 stage II patients (HR 4.45, p = 0.015) - fragile, state it. GSE72094 passes the registered rule (HR III vs I 3.09).
- Coverage: 26 independent evaluable pairs (CPTAC-3 15; Yao 2021 in GSE31210, GSE50081, GSE30219; GSE81089 8); GSE72094 none.
- H1 (vs expression-matched random genes, Holm within cohort): 0/26 Replicated, 0 Contradicted, 26 Inconclusive. Smallest Holm p: Yu 2024 GSE81089 0.076; Yao 2021 CPTAC-3 0.090. Unadjusted p < 0.05: Yao CPTAC-3 0.006, Yu GSE81089 0.009, Zhang 2021 CPTAC-3 0.014, Jin 2020 CPTAC-3 0.022, Wu 2022 CPTAC-3 0.024. Sensitive to the null pool: lncRNA-only pool -> Yu GSE81089 Replicated; all-gene pool -> Yao CPTAC-3 Replicated; stage I-II, early deaths, GSE31210-22: none.
- H2 (stage + sex + age, LRT, Holm): 6/26 add value - CPTAC-3 Li 2023 (p_holm 0.021), Wu 2022 (0.029), Yang 2022 (0.039); Yao 2021 GSE50081 (0.033) and GSE30219 (0.013); Yu 2024 GSE81089 (0.043). Delta C 0.03-0.06; bootstrap CI excludes 0 only for Wu 2022 CPTAC-3 (0.044, 0.0004-0.107).
- H3: C-index TCGA median 0.627 (0.609-0.705) -> CPTAC-3 0.591 (0.526-0.652) -> GSE81089 0.504 (0.480-0.624); TCGA minus CPTAC-3 median 0.059. Only Yao 2021 poolable: 5 cohorts, 232 deaths, HR 1.41 per SD, REML+HK 1.30-1.52, I2 0, tau2 0; leave-one-out 1.38-1.44. PH violated: CPTAC-3 4, GSE30219 1, GSE81089 1 (RMST reported for those, as registered).
- Direction consistency (Update 1): 23/25 effective observations positive (p = 1.9e-5); nominal 24/26. Max Jaccard 0.333 (Geng-Lin, one cluster); 14 clusters.
- Gene-level sign agreement in independent cohorts: 0.776.
- Transportability (not pre-registered, decision rule 11 Oct 2026): mean log-HR 0.243 (HR 1.27 per SD), bootstrap CI 0.127-0.370 -> "Transportable"; CR2 cross-check 0.251 (0.060-0.443, df 2.7); same verdict in all 7 sensitivity settings. By cohort: CPTAC-3 mean log-HR 0.287 (15 pairs), GSE81089 0.121 (8), GPL570 Yao 0.30-0.39.
- Power (C = 0.60 / 0.65 at alpha 0.05): CPTAC-3 0.61/0.94 (Holm step 0.32/0.74), GSE31210 0.48/0.82, GSE50081 0.64/0.92, GSE30219 0.53/0.87, GSE81089 0.54/0.92 (Holm 0.27/0.75), GSE72094 0.91/1.00 (no evaluable signature).

Issues found and fixes

1. Hartung-Knapp interval narrower than common-effect for Yao 2021 (q = Q/(k-1) = 0.19): HK 1.30-1.52 vs common effect 1.24-1.60 vs modified HK (q* = max(1, q)) 1.17-1.69. Fix: script 37_pooling_sensitivity.R (R/meta.R pool_ci_variants, test) reports common-effect, REML Wald and modified HK for full set and leave-one-out; keep registered HK as primary, report modified HK beside it (not pre-registered). Conclusion unchanged.
2. 108 v1 files sit in results/ (Table1-5, Figure1-4, Supplementary_Table_, *Yao2021_*, CPTAC3_, *robustness_*, _archive_old/, _pre_log2_fix/, paper_tables/), tracked in git - arrived 24 Oct 15:28 with the merge of the old GitHub repository (229fc8f). None is a v2 output; their numbers (e.g. Wu/Yao "Replicated", pooled 0.322 "Inconclusive") are superseded. Fix: check what else the merge brought (git diff --stat v2-preanalysis HEAD outside results; git ls-files), then git mv all v1 material to archive_v1/. Literature-level items (PRISMA counts, reporting completeness S22) to be re-verified, not reused blindly.
3. Story changed vs Draft 4: no signature Replicated (Draft 4: Wu 2022 and Yao 2021); aggregate now "Transportable" (Draft 4: Inconclusive, 11 pairs); Yao pooled over 5 cohorts. Title, abstract, results, discussion must be rewritten from v2 numbers only.
4. Interpretation: "Transportable" = average association in the published direction (vs HR = 1), not specificity; H1 says no single signature beats random genes; H2 "adds value" is relative to clinical covariates, not to random genes. Transport is cohort-dependent (GSE81089 median C 0.50).
5. Minor, report: C-index and HR disagree in direction for Yang 2022 and Lin 2024 in GSE81089 (C 0.493/0.497, HR 1.14/1.15) - rank vs magnitude near the null. RMST only where PH violated (registered); tau = smaller group's maximum follow-up (232 months in TCGA). 5 non-convergence warnings in script 34 bootstraps.

Next

- Run script 37; clean v1 files (after the git check).
- Script for v2 manuscript tables and figures (none of the existing Table/Figure files are v2).
- v2 deviations log (docs/): successor/predecessor IDs, overrides, D1-D6, 2 code fixes after freeze, pooling-interval sensitivity, transportability (not pre-registered).
- Draft 5 manuscript from results/numbers.csv and the v2 tables; decide title framing.